R&D | RetailPulse

Document ID: RP-EDA-007 | Version: 2.0 | Date: 2026-10-21 | Lane: Insights Analyst

Python Quick Charts (Optional)

> **In simple words**
>
> Python is optional here. We use it only for a few quick charts. All real work is in SQL and Power BI.

> **This document is optional**
>
> Use Python only for 2 or 3 quick charts and to look at the data once. Cleaning, baseline and analysis are done in SQL. Do not build models.

1. What Python is for

Use	Why it helps
Line chart of daily sales for 3 families	See seasonality and spikes before writing SQL
Histogram of sales for one family	See if many days are zero
Heatmap of WAPE (weekday x month) from SQL output	A nice image for the report

2. What not to do

- No forecasting models (ARIMA, Prophet, XGBoost, LSTM).
- No feature engineering or tuning.
- No cleaning that is not also done in SQL. The SQL scripts are the source of truth.

3. Libraries

- **pandas**: read the SQL output CSV files.
- **matplotlib / seaborn**: simple charts.

4. Output

One notebook `eda_charts.ipynb` in `06_Python_Optional` and 2 or 3 PNG images in `07_Screenshots`.

© 2026 R&D | RetailPulse - Internal Document